

Paper 2 - Set 06 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Exponentials and logarithms
2	D	Trigonometric functions and equations
3	B	Functions and transformations
4	E	Exponentials and logarithms
5	D	The logic of arguments
6	F	Sequences and series
7	A	Identifying errors in proofs
8	A	Mathematical proof
9	B	Angles, polygons, congruence and vectors
10	F	Integration
11	A	Identifying errors in proofs
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13	G	Ratio, proportion and percentages
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16	D	Chains of reasoning
17	H	Quadratic functions
18	D	Indices and surds
19	A	Identifying errors in proofs
20	A	The logic of arguments

Worked solutions

Question 1 LOG-01 · Exponentials and logarithms · spec MM5.2

Find the value of $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$.

Answer: A

$$\log_2 \left(\frac{40}{5} \cdot \frac{1}{4} \right) = \log_2 2 = 1.$$

Question 2 TRG-03 · Trigonometric functions and equations · spec MM4.3

Find the exact value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

Answer: D

$$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1.$$

Question 3 FUN-17 · Functions and transformations · spec MM8.3

The line $y = mx + c$ passes through $(2, 7)$. If m is increased by 1 and c is decreased by 2, the new line passes through $(2, k)$. Find k .

Answer: B

$$\text{New value at } x = 2: (m + 1) \cdot 2 + (c - 2) = 2m + c = 7. \text{ So } k = 7.$$

Question 4 LOG-16 · Exponentials and logarithms · spec MM5.1

For which of the following values of x is $3^x < 2^{x+1}$?

I $x = 0$

II $x = 1$

III $x = 2$

Answer: E

$$x = 0: 1 < 2 \text{ true. } x = 1: 3 < 4 \text{ true. } x = 2: 9 < 8 \text{ false. So I and II.}$$

Question 5 LGC-14 · The logic of arguments · spec Arg1

For a positive integer n , consider the statement: **if n is prime, then n is odd.** Which one of the following is true?

Answer: D

$$n = 2 \text{ is an even prime, so the statement is false. The converse, 'if } n \text{ is odd then } n \text{ is prime', fails for } n = 9.$$

Question 6 SEQ-12 · Sequences and series · spec MM2.3

Which of the following geometric series is/are convergent?

I $1 + 0.9 + 0.81 + \dots$

II $3 - 3 + 3 - 3 + \dots$

III $100 + 10 + 1 + 0.1 + \dots$

Answer: F

$$\text{A geometric series converges if and only if } |r| < 1. \text{ I has } r = 0.9, \text{ III has } r = 0.1: \text{ both converge. II has } r = -1 \text{ and does not converge.}$$

Question 7 ERR-18 · Identifying errors in proofs · spec Err2

A student solves $2\cos^2 x = \cos x$ for $0 \leq x < 2\pi$.

I Divide both sides by $\cos x$: $2\cos x = 1$.

II So $\cos x = \frac{1}{2}$.

III So $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$.

How many solutions has the student missed, and where is the first error?

Answer: A

Dividing by $\cos x$ loses $\cos x = 0$, i.e. $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$: two solutions.

Question 8 PRF-20 · Mathematical proof · spec Prf3

T_n denotes the n th triangular number $\frac{n(n+1)}{2}$. A student observes $T_1 + T_2 = 4$, $T_2 + T_3 = 9$, $T_3 + T_4 = 16$ and conjectures that $T_n + T_{n+1}$ is always a perfect square. Which of the following proves the conjecture?

Answer: A

Only A is a general algebraic verification.

Question 9 X2-20 · Angles, polygons, congruence and vectors · spec M5.3

$ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?

Answer: B

Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other: P is necessary. A non-rectangular parallelogram has bisecting diagonals without being a rectangle: not sufficient.

Question 10 INT-09 · Integration · spec MM7.1

Which of the following statements is/are true?

I $\int_{-2}^2 x^3 dx = 0$

II The area between $y = x^3$ and the x -axis from $x = -2$ to $x = 2$ is 0.

III $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$

Answer: F

I: odd function over a symmetric interval, so 0. II is false: the area is $2 \int_0^2 x^3 dx = 8$; the integral is 0 because the parts cancel. III: even function, true.

Question 11 ERR-03 · Identifying errors in proofs · spec Err2

A student tries to solve $\sin \theta = \sin 40^\circ$ for $0^\circ \leq \theta < 360^\circ$.

I Since $\sin \theta = \sin 40^\circ$, we have $\theta = 40^\circ$.

II The sine function has period 360° , so $\theta = 40^\circ + 360^\circ k$.

III In the given interval the only solution is $\theta = 40^\circ$.

Which of the following is correct?

Answer: A

'If $\sin \alpha = \sin \beta$ then $\alpha = \beta$ ' is a classic invalid deduction; $\sin(180^\circ - 40^\circ) = \sin 40^\circ$ too.

Question 12 LGC-10 · The logic of arguments · spec Arg2

For real numbers a and b , let P : ' $a = b$ ' and Q : ' $a^2 = b^2$ '. Which one of the following is true?

Answer: C

$a = b \Rightarrow a^2 = b^2$ (sufficient), but $a = 1, b = -1$ gives $a^2 = b^2$ without $a = b$ (not necessary).

Question 13 RAT-15 · Ratio, proportion and percentages · spec M3.8

Which of the following statements is/are true?

- I Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.
- II A 10% increase followed by a 10% increase is a 21% increase overall.
- III If A is 25% more than B , then B is 20% less than A .

Answer: G

I is false: $1.5 \times 0.5 = 0.75$. II: $1.1^2 = 1.21$, true. III: $A = 1.25B \Rightarrow B = 0.8A$, true.

Question 14 PRF-02 · Mathematical proof · spec Prf1

Which of the following is a counterexample to the claim '**if n is a positive integer, then $2^n + 1$ is prime**'?

Answer: D

$2^3 + 1 = 9 = 3 \times 3$ is not prime. ($n = 1, 2, 4, 8$ give 3, 5, 17, 257, all prime.)

Question 15 RSN-06 · Chains of reasoning · spec Prf5

Three statements are made about a positive integer N :

- N is a perfect square.
- N is a multiple of 6.
- N is less than 50.

Exactly two of the statements are true. Which of the following could NOT be N ?

Answer: C

36 makes all three statements true, not exactly two. Each of the others makes exactly two true: 16 and 25 (square, < 50), 48 and 12 (multiple of 6, < 50), 144 (square, multiple of 6).

Question 16 RSN-12 · Chains of reasoning · spec Prf5

How many positive integers less than 1000 are divisible by neither 2 nor 5?

Answer: D

Of 1 to 999: 499 multiples of 2, 199 of 5, 99 of 10. Divisible by 2 or 5: $499 + 199 - 99 = 599$. Neither: $999 - 599 = 400$.

Question 17 QUA-06 · Quadratic functions · spec MM1.3

The function $f(x) = x^2 - 6x + 13$ is defined for all real x . Which of the following statements is/are true?

- I $f(x) > 0$ for all real x
- II The graph of $y = f(x)$ is symmetric about the line $x = 3$
- III The equation $f(x) = 4$ has exactly one real solution

Answer: H

$f(x) = (x - 3)^2 + 4 \geq 4 > 0$, so I is true. The axis of symmetry is $x = 3$, so II is true. $f(x) = 4$ only when $(x - 3)^2 = 0$, i.e. $x = 3$: exactly one solution, so III is true.

Question 18 IND-12 · Indices and surds · spec MM1.1

For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?

Answer: D

For the sum to be rational, both $\sqrt{n}$ and $\sqrt{n+1}$ would need to be rational (if one is irrational, $\sqrt{n} + \sqrt{n+1}$ squared gives $2n + 1 + 2\sqrt{n(n+1)}$ and $n(n+1)$ is never a perfect square for $n \geq 1$ since $n^2 < n(n+1) < (n+1)^2$). No two consecutive positive integers are both squares, so the answer is 0.

Question 19 ERR-06 · Identifying errors in proofs · spec Err1

Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

- I $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$.
- II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.
- III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

Answer: A

f is (strictly) increasing because $f'(x) \geq 0$ with equality only at an isolated point, so $f(a) < f(b)$ whenever $a < b$; but line III's assertion ' $f'(x) > 0$ for all x ' is false at $x = 1$. There is no turning point at $x = 1$ (the derivative does not change sign).

Question 20 LGC-15 · The logic of arguments · spec Arg2

Let P : 'the integer n is divisible by 4' and Q : 'the integer n^2 is divisible by 8'. Which one of the following is true?

Answer: A

If $4 \mid n$ then $16 \mid n^2$, so $8 \mid n^2$: P is sufficient. Conversely, if $8 \mid n^2$ then n is even, say $n = 2m$, and $8 \mid 4m^2$ forces m to be even, so $4 \mid n$: P is also necessary. (Tempting wrong answer: 'sufficient but not necessary' - but no counterexample exists.)